1. Greater on right side
You are given an array Arr of size N. Replace every element with the next greatest element (greatest element on its right side) in the array. Also, since there is no element next to the last element, replace it with -1.
Input:
N = 6
Arr[] = {16, 17, 4, 3, 5, 2}
Output:
17 5 5 5 2 -1
Explanation: For 16 the greatest element
on its right is 17. For 17 it's 5.
For 4 it's 5. For 3 it's 5. For 5 it's 2.
For 2 it's -1(no element to its right).
So the answer is 17 5 5 5 2 -1
Link: https://www.geeksforgeeks.org/problems/greater-on-right-side4305/1

2. Given a Boolean matrix mat[M][N] of size M X N, modify it such that if a matrix cell mat[i][j] is 1 then make its adjacent cells as 0.
Input:
1 0 1
0 1 0
1 1 1
Output:
0 0 0
0 0 0
1 0 1
Explanation:
For the cell mat[0][0] which is 1, its adjacent cells (mat[0][1] and mat[1][0]) are modified to 0.
For the cell mat[1][1] which is 1, its adjacent cells (mat[0][1], mat[1][0], mat[1][2], and mat[2][1]) are modified to 0.
The modification is not applied to the cell mat[2][2] as it doesn't have all four adjacent cells.

3. Equilibrium index of an array is an index such that the sum of elements at lower indexes is equal to the sum of elements at higher indexes. For example, in an array A:
Example :
Input: A[] = {-7, 1, 5, 2, -4, 3, 0}
Output: 3
3 is an equilibrium index, because:
A[0] + A[1] + A[2] = A[4] + A[5] + A[6]
Input: A[] = {1, 2, 3}
Output: -1

4. In MS-Paint, when we take the brush to a pixel and click, the color of the region of that pixel is replaced with a new selected color. Following is the problem statement to do this task.
Given a 2D screen, location of a pixel in the screen and a color, replace color of the given pixel and all adjacent same colored pixels with the given color.
Example:
Input:
```
       screen[M][N] = {{1, 1, 1, 1, 1, 1, 1, 1},
                       {1, 1, 1, 1, 1, 1, 0, 0},
                       {1, 0, 0, 1, 1, 0, 1, 1},
                       {1, 2, 2, 2, 2, 0, 1, 0},
                       {1, 1, 1, 2, 2, 0, 1, 0},
                       {1, 1, 1, 2, 2, 2, 2, 0},
                       {1, 1, 1, 1, 1, 2, 1, 1},
                       {1, 1, 1, 1, 1, 2, 2, 1},
                       };
    x = 4, y = 4, newColor = 3
```
The values in the given 2D screen indicate colors of the pixels.

```
x and y are coordinates of the brush, newColor is the color that
should replace the previous color on screen[x][y] and all surrounding
pixels with same color.

Output:
Screen should be changed to following.
       screen[M][N] = {{1, 1, 1, 1, 1, 1, 1, 1},
                      {1, 1, 1, 1, 1, 1, 0, 0},
                      {1, 0, 0, 1, 1, 0, 1, 1},
                      {1, 3, 3, 3, 3, 0, 1, 0},
                      {1, 1, 1, 3, 3, 0, 1, 0},
                      {1, 1, 1, 3, 3, 3, 3, 0},
                      {1, 1, 1, 1, 1, 3, 1, 1},
                      {1, 1, 1, 1, 1, 3, 3, 1},
                      };
```

5. Given a matrix of 2D array of n rows and m coloumns. Print this matrix in ZIG-ZAG fashion as shown in figure.

Example:
Input:
1 2 3
4 5 6
7 8 9
Output:
1 2 4 7 5 3 6 8 9

6. Remove the duplicates in the String.
Testcase 1:
Input: Java1234
Output: Javb1234 (Remove the second ‘a’ as it is duplicated)
Testcase 2:
Input: Python1223:
Output: Python1234 (Replace the second 2 with 3, and replace 3 with 4 as 3 is replaced for the duplicated 2)
Testcase 3:
Input: aBuzZ9900
Output: aBuzC9012
(Replace the second ‘Z’ with ‘C’ as ‘a’ and ‘B’ are already there in the String. Replace with capital C as the letter to be replaced is capital Z. The second 9 turns out to be zero and the zero turns out to ‘1’ and the second zero turns out to ‘2’)

7. Print whether the version is upgraded, downgraded or not changed according to the input given.
example: Input : Version1 4.8.2 Version2 4.8.4 Output: upgraded, Input : Version1 4.0.2 Version2 4.8.4 Output: downgraded

8. Print all possible subsets of the given array whose sum equal to given N.
example: Input: {1, 2, 3, 4, 5} N=6 Output: {1, 2, 3}, {1, 5}, {2, 4}

9. Reverse the words in the given String1 from the first occurrence of String2 in String1 by maintaining white Spaces.
example: String1 = Input: This is a test String only String2 = st Output: This is a only String test

10. calculate Maximum number of chocolates can eat and Number of wrappers left in hand.
Money: Total money one has to spend.
Price: price per chocolate.
wrappers: minimum number of wrappers for exchange choco: number of chocolate for wrappers.
Max visit: Maximum number of times one can visit the shop.(if zero consider it infinite)

example: input: Money:40 Price:1 wrappers:3 choco:1 Max visit:1 Output: total chocolate can eat: 53 wrappers left in hand:14

11. Sample Input-

```
2
Hacker
Rank
Sample Output-
Hce akr
Rn ak
```

13. Print the word with odd letters – PROGRAM

Sample Output-+4477

```
P     P
 R   R
  O O
   G
  R R
 A   A
M     M
```

14. Sample Input – Alternate Sorting

Input: {1, 2, 3, 4, 5, 6, 7}

output: {7, 1, 6, 2, 5, 3, 4}

15. Given an array of values persons[], each represents the weight of the persons. There will be infinite bikes available. Given a value K which represents the maximum weight that a bike accommodates. Along with that one more condition, a bike can carry two persons at a time. You need to find out the least number of times, the bike trips are made.

16. Assume there exists infinite grid, you're given initial position x, y. Inputs will be movements either L or R or U or D. After n inputs, you need to give the current position.

•Input:
•4 5 //initial position x, y
•9 //number of movements
•U L R R D D U L R //7 movements
•Output:
5 5

•Given a matrix NxN, you are initially in the 0, 0 position. The matrix is filled with ones and zeros. Value "one" represents the path is available, while "zero" represents the wall. You need to find the can you able to reach the (N-1)x(N-1) index in the matrix. You can move only along the right and down directions if there's "one" available.

•Input:
•5 //N value
•1 0 1 0 0
•1 1 1 1 1
•0 0 0 1 0
•1 0 1 1 1
•0 1 1 0 1
•Output:
Yes

17. Given an array of integers, compute the maximum value for each integer in the index, by either summing all the digits or multiplying all the digits. (Choose which operation gives the maximum value)

•Input:
•5
•120 24 71 10 59
•Output:
•3 8 8 1 45

Explanation: For index 0, the integer is 120. Summing the digits will give 3, and whereas Multiplying the digits gives 0. Thus, maximum of this two is 3.

18. -1 represents ocean and 1 represents land find the number of islands in the given matrix.

```
Input:   n*n matrix
       1 -1 -1  1
      -1  1 -1  1
      -1 -1  1 -1
      -1 -1 -1  1
```

Output: 2 (two islands that I have bold in matrix at 1, 1 and 2, 2)

19. Print all the possible subsets of array which adds up to give a sum.

```
Input: array{2, 3, 5, 8, 10}
       sum=10
Output: {2, 3, 5}
       {2, 8}
       {10}
```

20. There is a circular queue of processes. Every time there will be certain no of process skipped and a particular start position. Find the safe position.

```
Input: Number of process:5
       Start position:3
       Skip: 2nd
Output: 1 will be the safest position
(Logic: 1 2 3 4 5 starting from 3, 5th process will be skipped
        1 2 3 4 5 process 2 will be skipped
        1 2 3 4 5 process 4 will be skipped
        1 2 3 4 5 process 3 will be skipped, so safest process is 1.
```

21. Given N. print the following snake pattern (say N = 4). condition: must not use arrays (1D array or 2D array like Matrix).

```
1   2   3   4
8   7   6   5
9   10  11  12
16  15  14  13
```

22. Given N. print the Latin Matrix (say N = 3). condition: must not use strings(aka character literals), arrays (both 1D and 2D), inbuilt functions(like rotate).

```
A   B   C
B   C   A
C   A   B
```

23. Given a number N. find the minimum count of numbers in which N can be represented as a sum of numbers x1, x2, … xn. where xi is number whose digits are 0s and 1s.

example 1) i/p : N = 33
o/p : count = 3.   33(11 + 11 + 11)
some other possibilities of 33 is (11 + 11 + 10 + 1),   (11 + 10 + 10 + 1 + 1), (10 + 10 + 10 + 1 + 1 + 1)

24.   Finding all permutations of a string. (backtracking approach).

25.   Given an array of integers, write a program to re-arrange the array in the given form.
1st_largest, 1st_smallest, 2nd_largest, 2nd_smallest, 3rd_largest ……. etc.

26. Sort the given elements in decending order based on the number of factors of each element – Solution 1

27. Find whether the given number is palindrome or not. Don’t use arrays or strings

28. Reverse the given string keeping the position of special characters intact

29. Find the shortest path from one element to another element in a matrix using right and down moves alone. The attached solution uses moves in all directions. – Solution 4

30. Pattern

31. Pangram Checking
Check whether all english alphabets are present in the given sentence or not
I/P: abc defGhi JklmnOP QRStuv wxyz
O/P: True
I/P: abc defGhi JklmnOP QRStuv
O/P: False

32. Password Strength
Find the strength of the given password string based on the conditions
Four rules were given based on the type and no. of characters in the string.
Weak – only Rule 1 is satisfied or Rule 1 is not satisfied
Medium – Two rules are satisfied
Good – Three rules satisfied
Strong – All Four rules satisfied

I/P: Qw!1   O/P: Weak
I/P: Qwertyuiop O/P: Medium
I/P: QwertY123   O/P: Good
I/P: Qwerty@123   O/P: Strong

33. First Occurrences
Given two strings, find the first occurrence of all characters of second string in the first string and
print the characters between the least and the highest index
I/P: ZOHOCORPORATION PORT
O/P: OHOCORPORAT
Explanation: The index of P in first string is 7, O is 1, R is 6 and T is 11. The largest range is 1 – 11.
So print the characters of the first string in this inex range i.e. OHOCORPORAT

34. Matrix Diagonal sum
Given a matrix print the largest of the sums of the two triangles split by diagonal from top right to bottom left
I/P:
3 3
1 2 3
4 5 6
7 8 9
O/P: 38

35. Matrix Addition
Given n integer arrays of different size, find the addititon of numbers represented by the arrays
I/P: 4
3 5 4 2
2 4 5
4 5 6 7 8
4 9 2 1
1 2
O/P: 50856

36. Many students will able to solve 3 problems in this round. So make sure you stand apart from the crowd.Their vacancy is going to be 5 for a team. The performance in this round could be taken as a tie breaker for round 3.
input : aaabbcc
output : abc

37. Evaluate the expression and sort and print the output. Getting the input is the tricky part
Input:
Number of input : 4
2*3
2^2^2
35
3*1
Output:
3*1
2*3
2^2^2
35

38. Given a 6 blocks, of different height h1, …, h6 . Make 2 towers using 3 Blocks for each tower in desired height h1, h2. Print the blocks to be used in ascending order
Input:
1 2 5 4 3 6
height of tower: 6 15
Output :